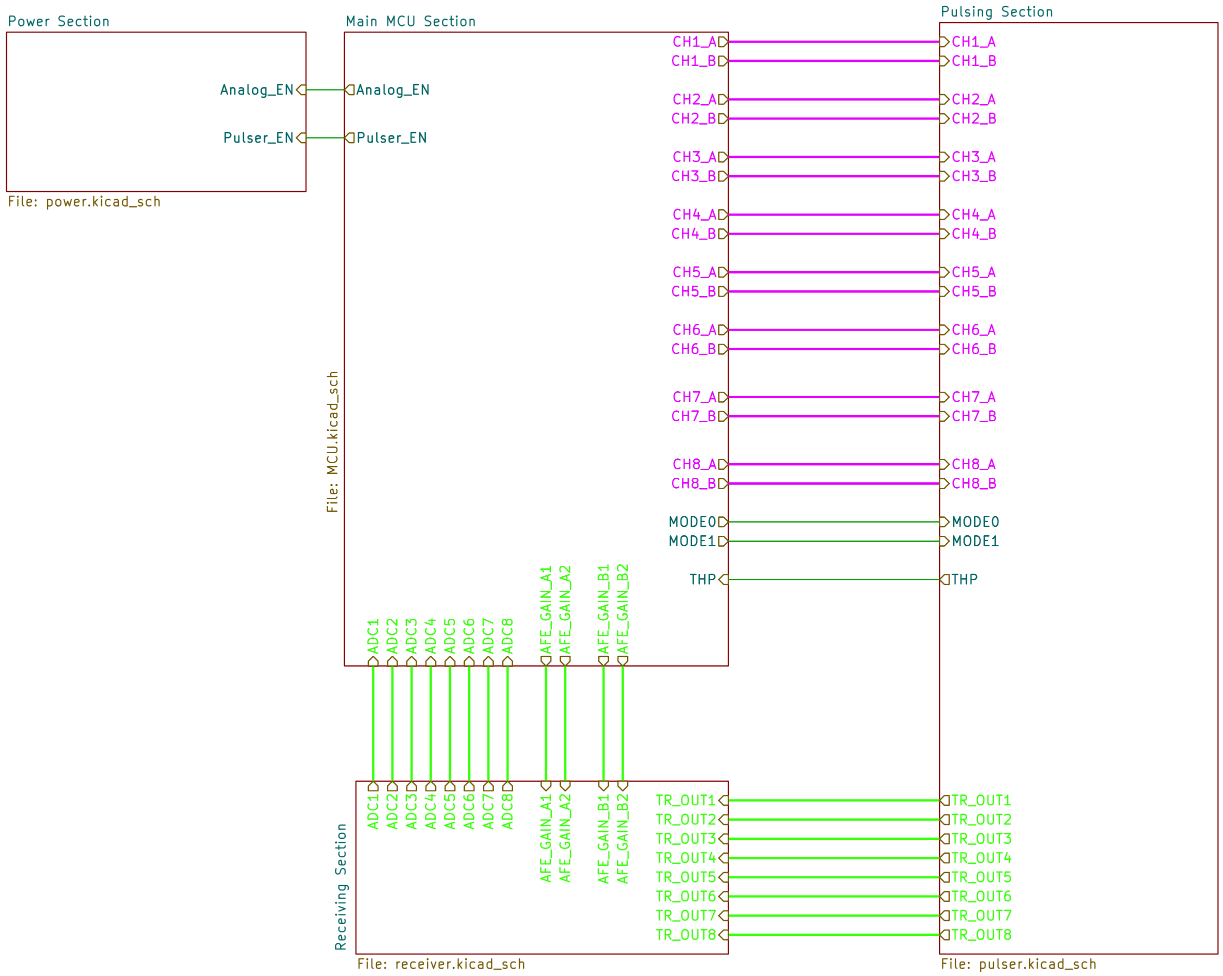


<table>
<tr><td rowspan="6">12 Volt
(External)
(555mA)</td><td rowspan="3">7 Volt
(Buck Converter)
(350mA)</td><td rowspan="2">5 Volt
(LDO Regulator)
(200mA)</td><td>LNA + VGA Supply
(160mA – short circuit)</td></tr>
<tr><td>Op-Amps
(4 * 10mA = 40mA)</td></tr>
<tr><td>3.3 Volt
(LDO Regulator)
(150mA)</td><td>MCU Supply
(150mA)</td></tr>
<tr><td rowspan="3">5 Volt
(Buck Converter)
(205mA)</td><td>Positive Pulse Driver Supply
(100mA)</td><td></td></tr>
<tr><td>3.3 Volt
(LDO Regulator)
(5mA)</td><td>Pulser Logic Supply
(5mA)</td></tr>
<tr><td>-5 Volt
(Inverting Buck Converter)
(100mA)</td><td>Negative Pulse Driver Supply
(100mA)</td></tr>
</table>

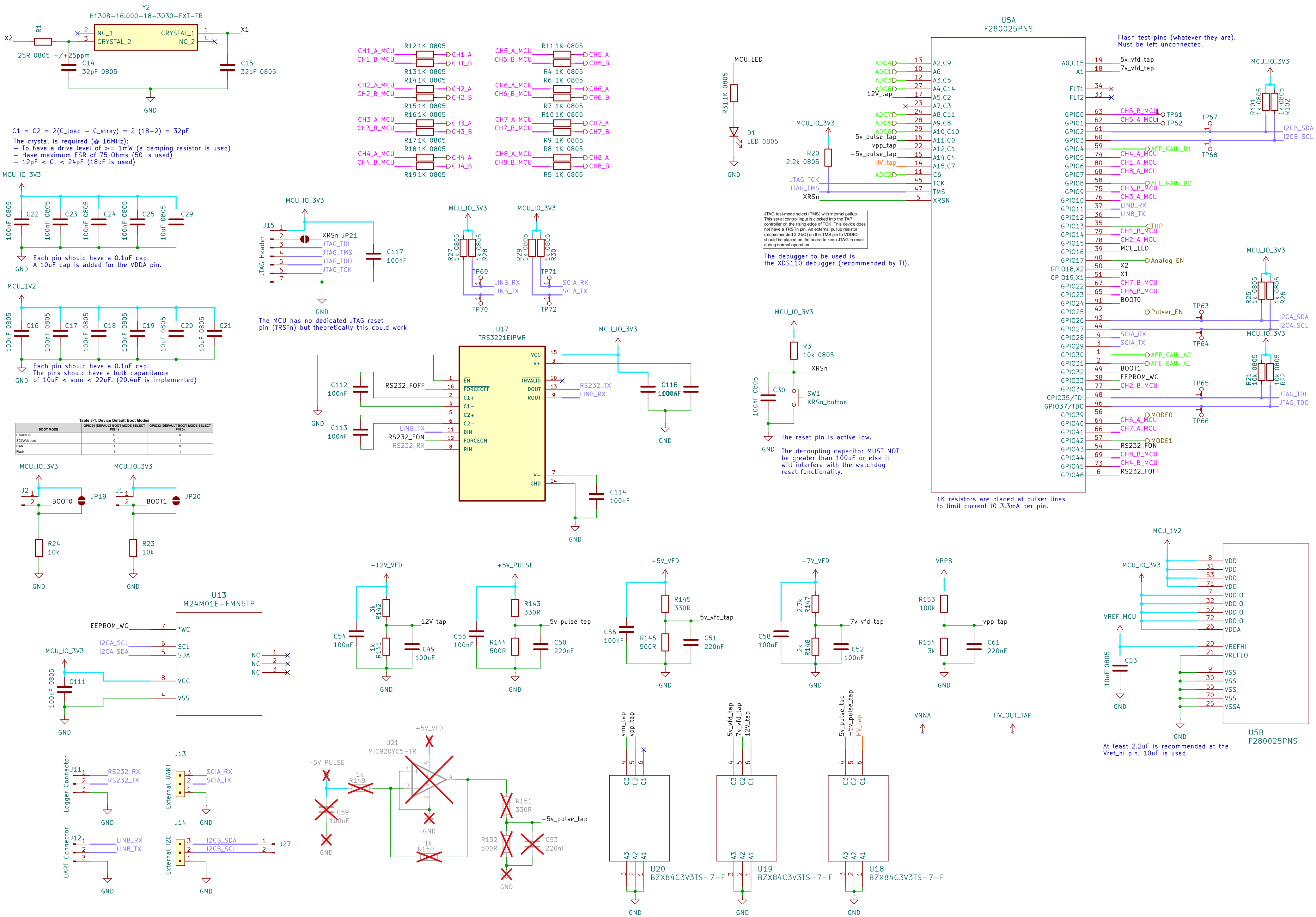
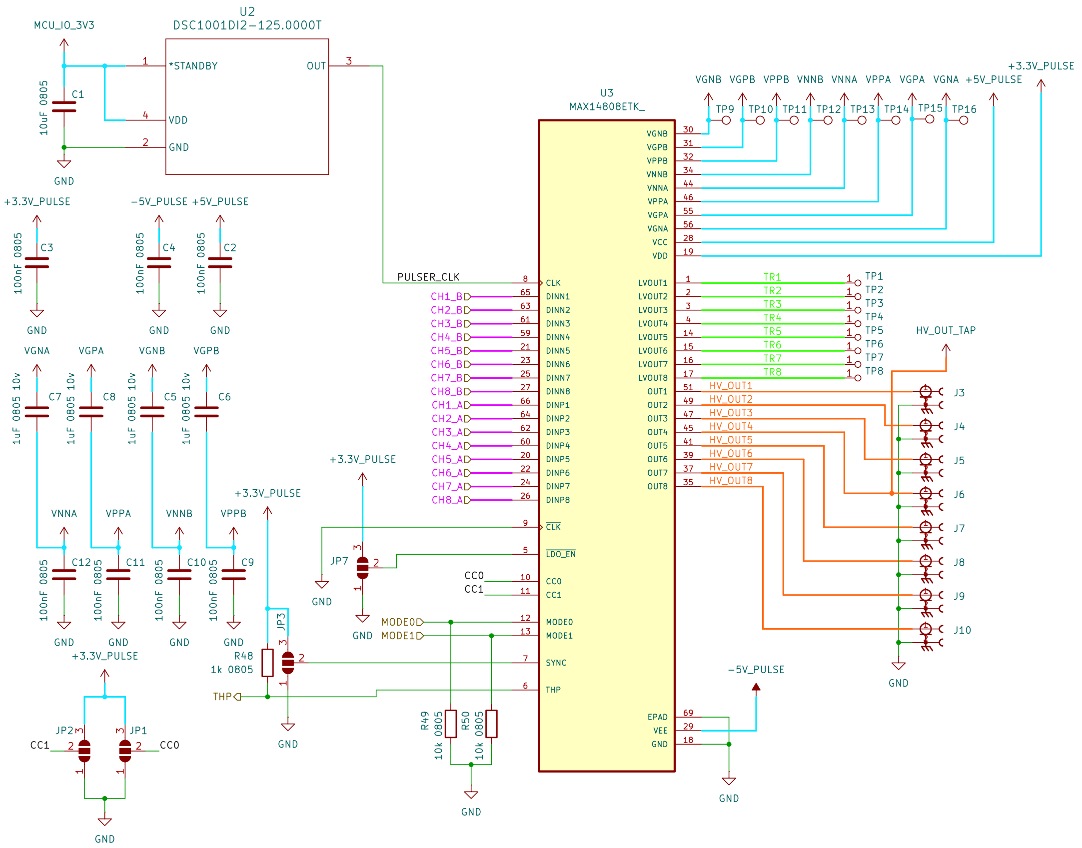
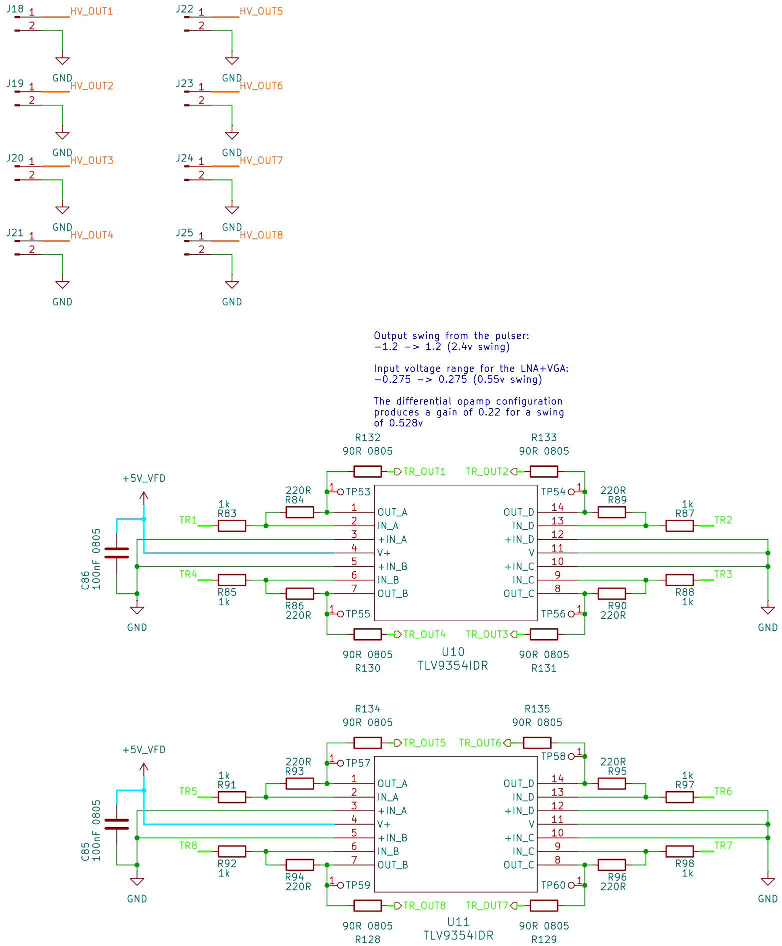


Table 3-1. Device Default Boot Modes

BOOT MODE	GPIO24 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI/Wait boot	0	1
CAN	1	0
Flash	1	1





VGXX: gate driver voltage supplies
(positive and negative for ports A & B)

VNNX/VPPX: positive and negative HV supplies
(for ports A & B)

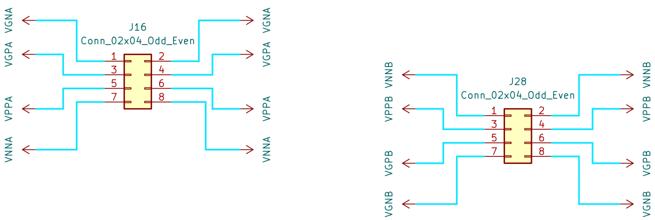
Enabling/Disabling the internal floating power supplies (FPS):

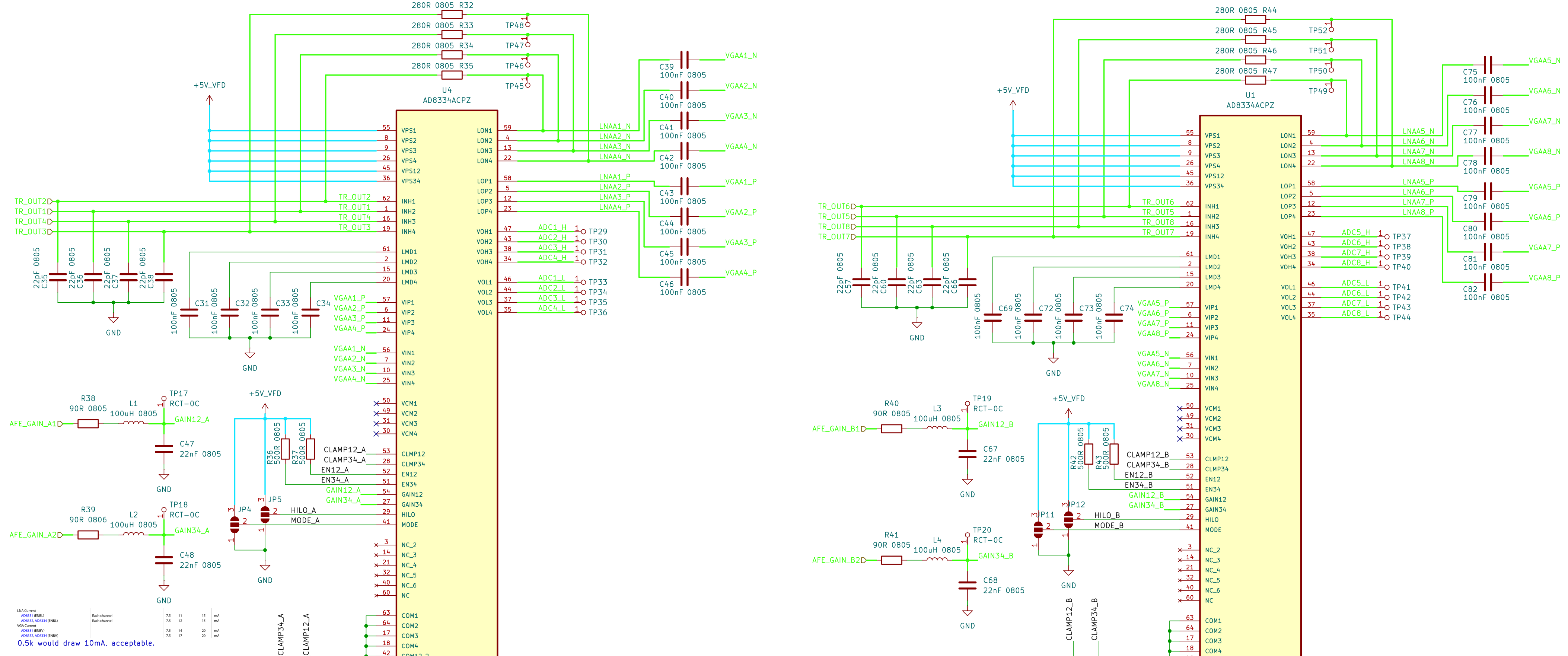
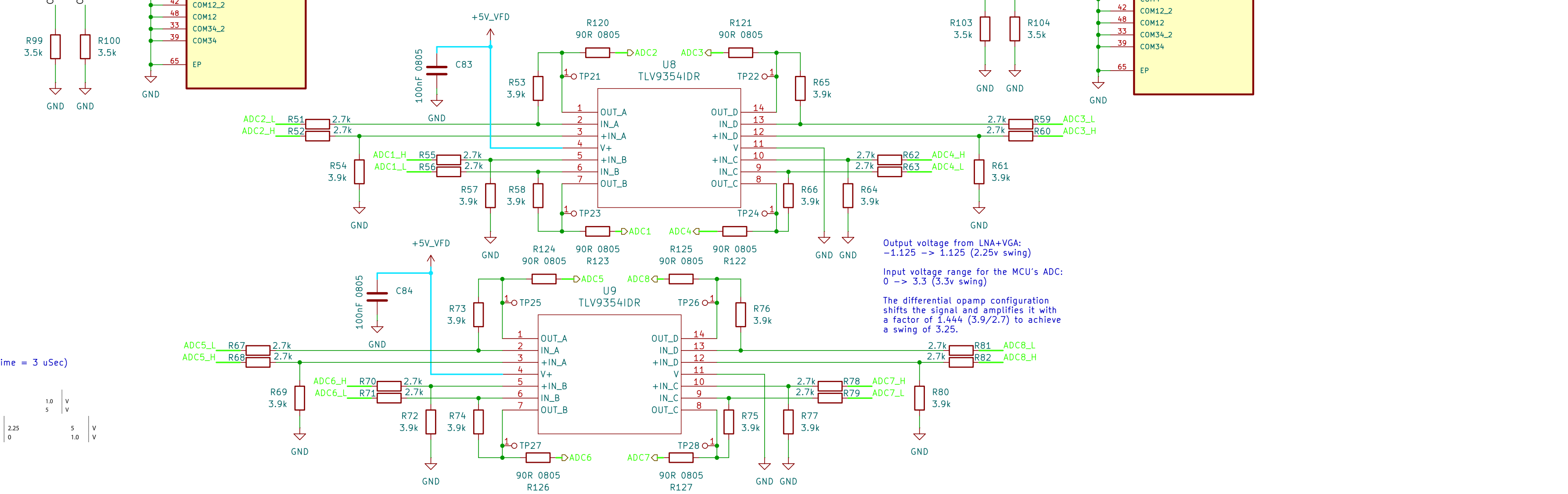
requirements in most of the cases. Drive $\overline{\text{LDO\_EN}}$ low or leave unconnected to enable the internal FPSs; drive $\overline{\text{LDO\_EN}}$ high to disable the internal FPSs.

A triple solder jumper is placed
to either pull-up (select external FPS)
or pull-down (select internal FPS) the
LDOEN pin.

Table 5. Current Drive Selection

INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A



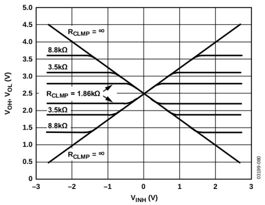


Figure 80. Output Clamping Characteristics

LNA Current					
AD8331 (ENBL)	Each channel	7.5	11	15	mA
AD8332, AD8334 (ENBL)	Each channel	7.5	12	15	mA
VGA Current					
AD8331 (ENBV)		7.5	14	20	mA
AD8332, AD8334 (ENBV)		7.5	17	20	mA

0.5k would draw 10mA, acceptable.

Table 7. LNA External Component Values for Common Source Impedances

R_{IN} (Ω)	R_{IZ} (Nearest STD 1% Value, Ω)	C_{SH} (pF)
50	280	22
75	412	12
100	562	8
200	1.13 k	1.2
500	3.01 k	None
6 k	∞	None

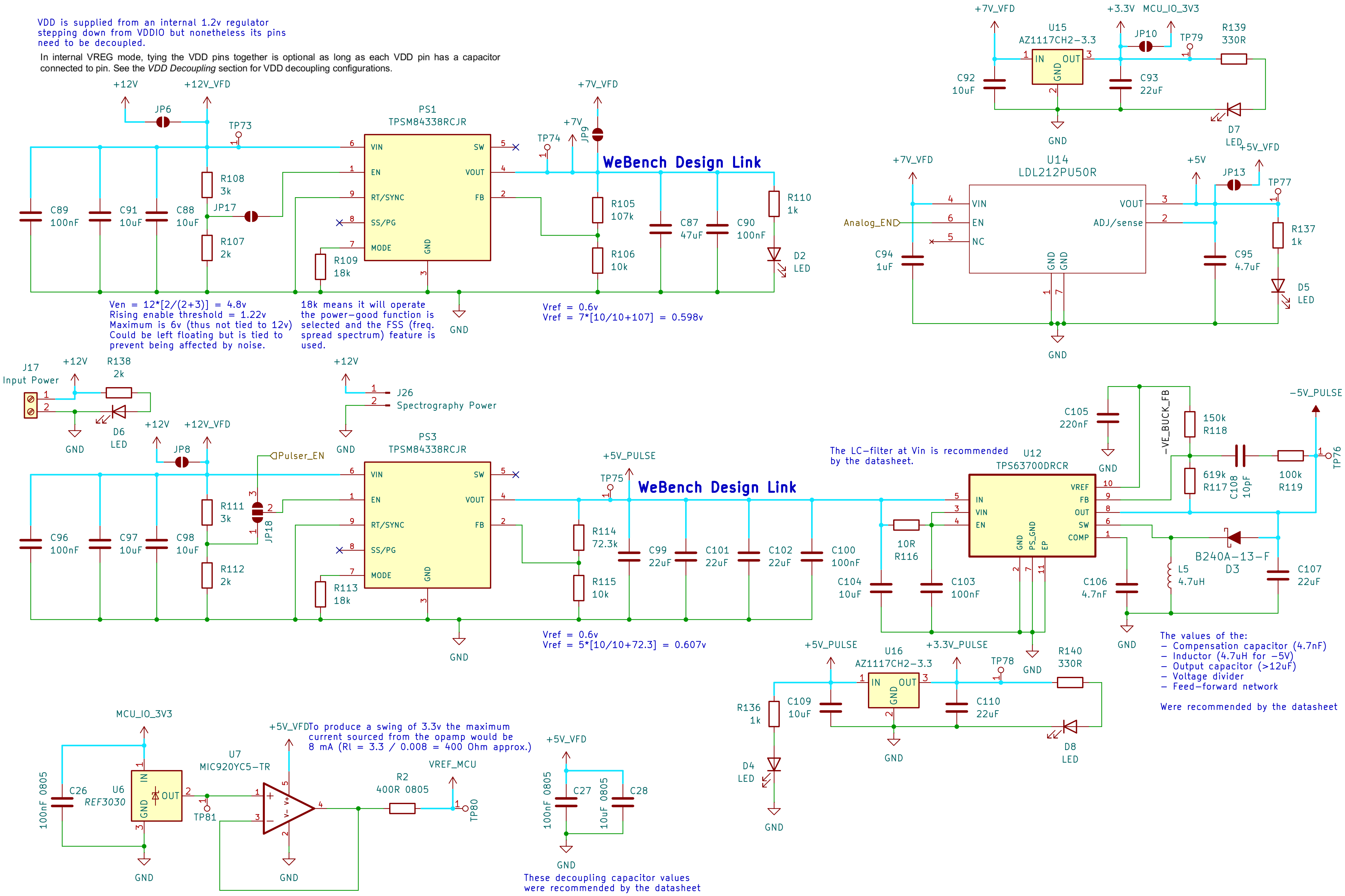
2nd order filter values were recommended
by a TI application note (spraa88a), specs:
– PWM frequency = 5 MHz
– Cut–off frequency = 107.3 KHz
– Damping ratio = 0.675 (over–damped but response time = 3 uSec)

MODE INTERFACE (PIN MODE)			
Logic Level for Positive Gain Slope	0	1.0	V
Logic Level for Negative Gain Slope	2.25	5	V
HILO GAIN RANGE INTERFACE (PIN HILO)			
Logic Level to Select HI Gain Range	2.25	5	V
Logic Level to Select LO Gain Range	0	1.0	V

Output voltage from LNA+VGA:
–1.125 –> 1.125 (2.25v swing)

Input voltage range for the MCU's ADC:
0 –> 3.3 (3.3v swing)

The differential opamp configuration
shifts the signal and amplifies it with
a factor of 1.444 (3.9/2.7) to achieve
a swing of 3.25.


VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.
In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the VDD Decoupling section for VDD decoupling configurations.
+12V
+12V_VFD
JP6
TP73
PS1
TPSM84338RCJR
VIN
EN
RT/SYNC
SS/PG
MODE
GND
SW
VOUT
FB
+7V_VFD
+7V
TP74
JP9
WeBench Design Link
C89
100nF
C91
10uF
C88
10uF
R108
3k
JP17
R107
2k
R109
18k
R105
107k
R106
10k
C87
47uF
C90
100nF
R110
1k
D2
LED
GND
Ven = 12*[2/(2+3)] = 4.8v
Rising enable threshold = 1.22v
Maximum is 6v (thus not tied to 12v)
Could be left floating but is tied to prevent being affected by noise.
18k means it will operate the power-good function is selected and the FSS (freq. spread spectrum) feature is used.
Vref = 0.6v
Vref = 7*[10/10+107] = 0.598v
U15
AZ1117CH2-3.3
IN
OUT
+3.3V
MCU_IO_3V3
JP10
TP79
R139
330R
C92
10uF
C93
22uF
D7
LED
U14
LDL212PU50R
Analog_EN
VIN
EN
NC
VOUT
ADJ/sense
+5V
JP13
+5V_VFD
TP77
R137
1k
C94
1uF
C95
4.7uF
D5
LED
J17
Input Power
R138
2k
D6
LED
+12V
J26
Spectrography Power
JP8
Pulser_EN
PS3
TPSM84338RCJR
JP18
R111
3k
R112
2k
R113
18k
C96
100nF
C97
10uF
C98
10uF
+5V_PULSE
TP75
R114
72.3k
R115
10k
C99
22uF
C101
22uF
C102
22uF
C100
100nF
Vref = 0.6v
Vref = 5*[10/10+72.3] = 0.607v
C105
220nF
-VE_BUCK_FB
150k
R118
619k
R117
C108
10pF
100k
R119
-5V_PULSE
TP76
The LC-filter at Vin is recommended by the datasheet.
U12
TPS63700DRCR
VREF
FB
OUT
SW
COMP
PS_GND
EP
10R
R116
C104
10uF
C103
100nF
C106
4.7nF
L5
4.7uH
B240A-13-F
D3
C107
22uF
The values of the:
- Compensation capacitor (4.7nF)
- Inductor (4.7uH for -5V)
- Output capacitor (>12uF)
- Voltage divider
- Feed-forward network
Were recommended by the datasheet
+5V_PULSE
U16
AZ1117CH2-3.3
+3.3V_PULSE
TP78
R140
330R
R136
1k
C109
10uF
C110
22uF
D4
LED
D8
LED
MCU_IO_3V3
C26
100nF 0805
U6
REF3030
TP81
+5V_VFD
U7
MIC920YC5-TR
To produce a swing of 3.3v the maximum current sourced from the opamp would be 8 mA (Rl = 3.3 / 0.008 = 400 Ohm approx.)
R2
400R 0805
VREF_MCU
TP80
+5V_VFD
C27
100nF 0805
C28
10uF 0805
These decoupling capacitor values were recommended by the datasheet